

Article

Integration and Validation of Soft Wearable Robotic Gloves for Sensorimotor Rehabilitation of Human Hand Function

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Abstract: This study aims to present the development of a wearable prototype device consisting of soft robotic gloves (SRGs), its integration into a wearable robotics platform for sensorimotor rehabilitation, and the device's validation experiments with individuals suffering from impaired hand motor function due to neurological lesions. The SRG is tested and evaluated by users with spinal cord injury (SCI) and stroke. The proposed system combines multiple-sensor arrays with pneumatic actuation to assist finger movement during grasping tasks. Evaluations on SCI and stroke patients revealed that the gloves consistently improved finger and grip performance. Detailed analyses indicated observable differences in sensor-derived features during actuation versus non-actuation, with statistically significant modifications appearing in both time-domain and frequency-domain metrics. Although the stroke participants exhibited greater variability, all participants were able to use the system reporting low discomfort and effort. The findings underscore the potential for personalized calibration to further optimize therapeutic outcomes. In summary, the study validates the utility of these gloves as assistive and rehabilitative modalities, and future research will focus on refining the device in the context of multimodal wearable robotics and individualized neurorehabilitation strategies.

Keywords: soft robotics; wearable robotics; robotic gloves; neurorehabilitation; spinal cord injury; stroke



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1. Introduction

The human hand's intrinsic biomechanical complexity and its overlying multi-level neuronal integration [1] enables precise voluntary movement [2,3], allows for detailed interaction with the surrounding world [4], plays a crucial role in multisensory integration in the brain [5], and is integral in the execution of many activities of daily living (ADLs) [6]. Loss of hand function is a traumatic consequence of many neurological conditions [7],

including SCI and stroke, that significantly affects patient quality of life (QoL) [8,9], acting as a major barrier to patient independence [10,11].

Over 15 million people worldwide live with spinal cord injury—most resulting from preventable trauma such as falls, traffic collisions, or violence—and younger age groups are disproportionately affected [12]. Stroke afflicts more than 100 million individuals globally, with over 15 million new cases each year, and remains the third leading cause of death and disability, accounting for some 160 million disability-adjusted life years lost [13,14]. Both SCI and stroke survivors face elevated risks of secondary complications, increased morbidity and premature mortality, and reduced educational and economic participation. Addressing this dual burden demands integrated strategies for prevention, acute care, rehabilitation, and long-term support to enhance outcomes and societal well-being [15].

Rehabilitation and restoration of hand function after neural damage can be challenging due to the aforementioned biomechanical and neural complexity. Therefore, addressing the challenge requires innovative strategies, among which lies the vision of creating robotic systems that mimic the functionality of human hand physiology [16]. In this context, soft robotics (SR) presents a promising solution by offering improved adaptability and harmony with the hand's dynamic movements [17]. Exercise-based treatments that attempt to stimulate affected muscles' activation while training the patient to perform specific activities have improved the neurologic recovery of the upper extremity in SCI and stroke [18]. The success of these treatments warrants further exploration of their relevance to hand function [19]. In addition, SCI and stroke survivors display a diverse range of phenotypes from complete paralysis to partial weakness and varied degrees of spasticity [20]. Although many patients retain some degree of hand function after SCI or stroke, performing ADLs [21] remains a significant challenge. Achieving complete recovery is often difficult and, in many cases, impossible [22].

To enhance rehabilitation—even in cases of partial recovery—it is essential to employ targeted motor exercises that both quantify patient progress and gauge therapeutic efficacy [23]. Conventional physical therapy is labor-intensive and costly [24], motivating the development of portable, user-friendly orthotic and robotic devices that simplify donning and doffing while remaining affordable [25]. Soft robotic gloves (SRGs) leverage compliant materials and advanced control systems to augment hand function and support sensorimotor retraining [26], yet optimal protocols for their clinical integration remain undefined [27]. We hypothesize that embedding SRGs within neurorehabilitation regimens will enhance motor recovery and drive neuroplastic changes, ultimately improving outcomes for individuals with hand deficits due to SCI or stroke. This study therefore aims to identify and evaluate strategies for effective SRG implementation in therapeutic settings, thereby maximizing their rehabilitative impact and quality-of-life benefits.

In this manuscript, we present the design, development, and evaluation experiments integrating ADL features of a wearable prototype device consisting of soft robotic gloves (SRGs). The SRG is tested and evaluated by users with spinal cord injury (SCI) and stroke with impaired hand function. Data from a multiple-sensor array are utilized to drive pneumatic actuation to assist finger movement during grasping tasks. Analyses are performed on physiological signals to evaluate actuated against non-actuated performance by users, while the user experience and the perception of robotics are also presented.

2. Materials and Methods

2.1. Body-Area-Distributed Sensor and Actuator System and the NeuroSuitUp Platform

The SRG system was developed as part of a larger assistive wearable robotics suite, designed to enable neuroplasticity-induced neurorehabilitation for patients post-spinal cord injury and stroke (Figure 1). The platform records the kinematic and dynamic response

of users during serious game-mediated training regimens, and generates the corresponding control response, through the use of lightweight, flexible, and dedicated actuation techniques [28]. The complementary wearable modalities of the suite include an upper and lower-body cloth-based exoskeleton, ensuring ease of use and minimal impact on the participants while providing a multi-segmental view of movement. Kinematic data are provided by 9 degrees of freedom (DoF) inertial measurement units (IMUs) placed on the primary kinematic segments of the corresponding body region (5 for the upper body, 4 for the lower body) to enable real-time recording of motion. In order to evaluate the underlying dynamics of the motor system, surface electromyography (sEMG) units are employed in the major muscle groups actuating the corresponding monitored kinematic joints (14 for the upper, 4 for the lower body).

Given the extensive impact of these pathologies on movement, the modalities are equipped with an electrical muscle stimulation (EMS) subsystem, providing assistance when the desired range-of-motion and ambient force generation capabilities of each user are insufficient for the task. The control strategies focus on providing stimulation in response to accumulated kinematic errors, specifically when participants fail to achieve target joint angles (position errors) or desired movement velocities (velocity errors) during task execution [29].

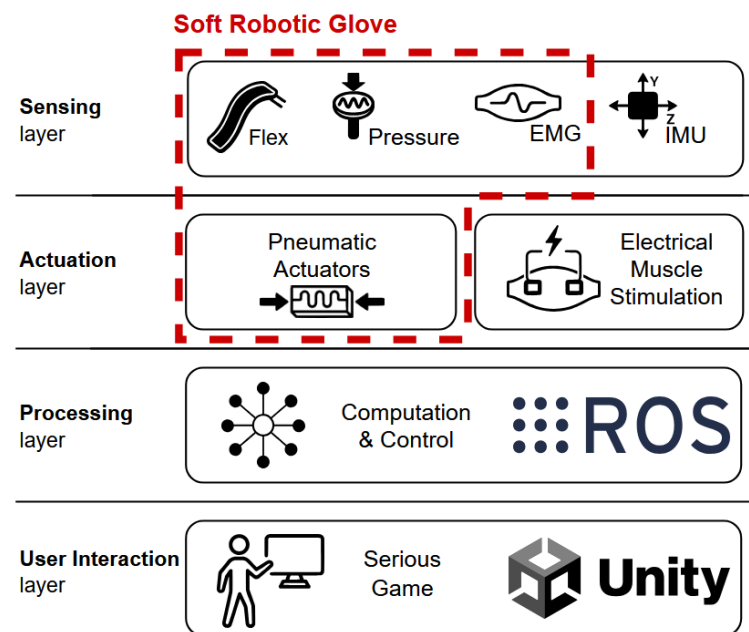


Figure 1. NeuroSuitUp - HEROES suite architecture.

Sensor data acquisition, processing, and control strategies employed by the suite are managed through a dedicated set of robot operating system (ROS) packages. This modular ROS stack enables reliable communication between operational nodes, including sensors, data processing filters, and control algorithms, whilst facilitating experimentation with different control strategies, under repeatable conditions. The processed sensor data are subsequently integrated into a custom serious game application, developed in Unity. The platform delivers interactive task-based training regimens, designed by healthcare professionals after a comprehensive physical evaluation, to ensure full adaptability to the patient's functional status and rehabilitation goals [29]. The continuous interaction between sensor data acquisition, real-time visual feedback through the serious gaming application, and the control actuation strategies aims to establish a robust biofeedback loop. This loop

is designed to promote neuroplasticity and enable a highly personalized, motivating, and immersive neurorehabilitation program.

The Wearable SRG

Wearable SRGs have emerged as a game-changing invention in the rehabilitation of people with hand impairments, providing new opportunities for regaining hand function and improving overall QoL [30]. These devices are intended to help patients undertake therapeutic regimens while also fostering brain neuroplasticity and motor skill development via interactive, user-friendly interfaces. The wearable SRGs presented in this manuscript incorporate modern technologies such as sensors, SR actuators, and adaptive control software systems, which can give individualized assistance and feedback during rehabilitation sessions.

2.2. Design, Development, and Integration of SRGs

The SRGs presented in this manuscript leverage a combination of multi-sensor data and SR exoskeleton actuation for sensorimotor rehabilitation of the human hand (Table 1) [31]. The emphasis on wearability, portability, and power autonomy is key to this effort, as it aligns with the practical needs of rehabilitation settings and the requirement for user-friendly therapies. The proposed system consists of three interrelated subsystems designed to synergistically facilitate rehabilitation and produce the best possible therapeutic outcomes: (a) The multi-sensor subsystem (MSS) encompasses a suite of sensors strategically integrated into the SRG, enabling data acquisition essential for monitoring hand movements and muscle activity for assessing the user's hand motor performance and physiological responses. Complementing this sensing capability is (b) the soft robotic pneumatic actuation subsystem (SRPAS), featuring a PneuNet-based [32] flexible soft-shell robotic exoskeleton engineered to apply targeted forces and stimuli to the user's hand [33]. Crucially, the SRPAS operates in tandem with a (c) microcontroller-based Communication Software Subsystem (ComSS), orchestrating near-real-time processing of the sensor data collected by the MSS, enabling precise control over the SRPAS movements and interactions with the user's hand. This approach has the potential to advance the area of neurorehabilitation by providing a dynamic and adaptable platform for tailored therapy interventions aimed at restoring and improving hand motor function.

Table 1. SRG system architecture.

Subsystem	Components	Function	Interaction
Multi-sensor subsystem (MSS)	Piezoelectric pressure sensor, finger flexion sensors (across fingers), sEMG sensor, accelerometer	Measures piezoelectric pressure, finger flexion, muscle activity, and hand 3D position. Collects data on hand movements and muscle responses	Sends sensory data to communication software subsystem (ComSS) for processing. Data used to evaluate motor performance and physiological responses.
Soft robotic pneumatic actuation subsystem (SRPAS)	PneuNet-based flexible soft robotic exoskeleton	Applies controlled forces to the hand for therapy and motor function improvement	Receives control signals from ComSS.
Communication software subsystem (ComSS)	Microcontroller-based, control algorithms. Data collection software	Processes real-time sensor data. Sends commands to the SRPAS	Orchestrates the interaction between MSS and SRPAS. Ensures real-time response to hand movements and therapy stimuli.

2.2.1. SRG Prototype for SCI Participants

The initial prototype was designed and developed for the right hand of individuals with SCI. This SRG is designed to monitor, evaluate, and facilitate the challenges faced by SCI patients, including reduced muscle control and sensory deficits. A Qwiic Flex Glove Controller [34] attached to the index and middle fingers is connected with the Qwiic Flex Glove Controller attached to the ring and little fingers. The Qwiic Flex Glove Controller is connected to the microcontroller via the SparkFun Qwiic Shield's Qwiic connection ports. One of the leads of each pressure sensor is connected to the power (3.3 V) and the other leads to the analog in of the microcontroller while a 10 kOhm pulldown resistor is connected between GND and the analog in. An sEMG Myoware Muscle Sensor's (+) lead is connected to the power (3.3 V), the (−) lead is connected to ground and the signal lead is connected to the analog in of the microcontroller. In addition to the MSS suite, the system incorporates the SRPAS subsystem based on a PneuNet [32] flexible exoskeleton that applies controlled forces to the hand. The sensor and actuation technologies involved are sewn, attached and wired on the surface of an F.F. Group Antistatic Glove [35] as illustrated in Figure 2.

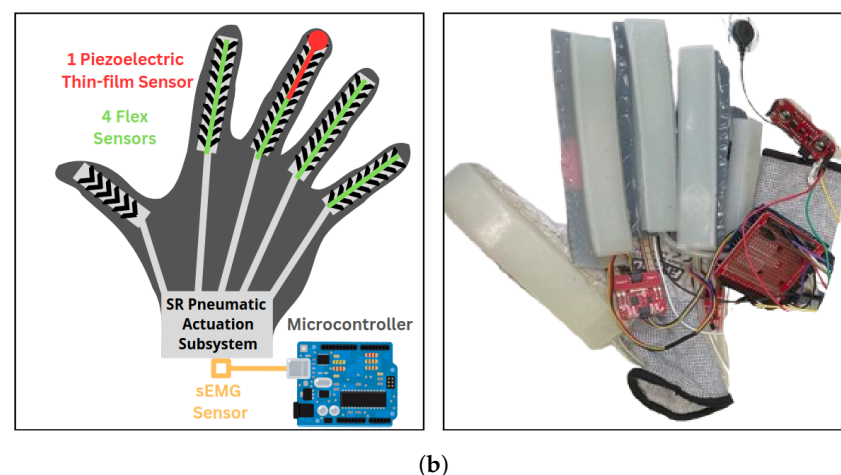
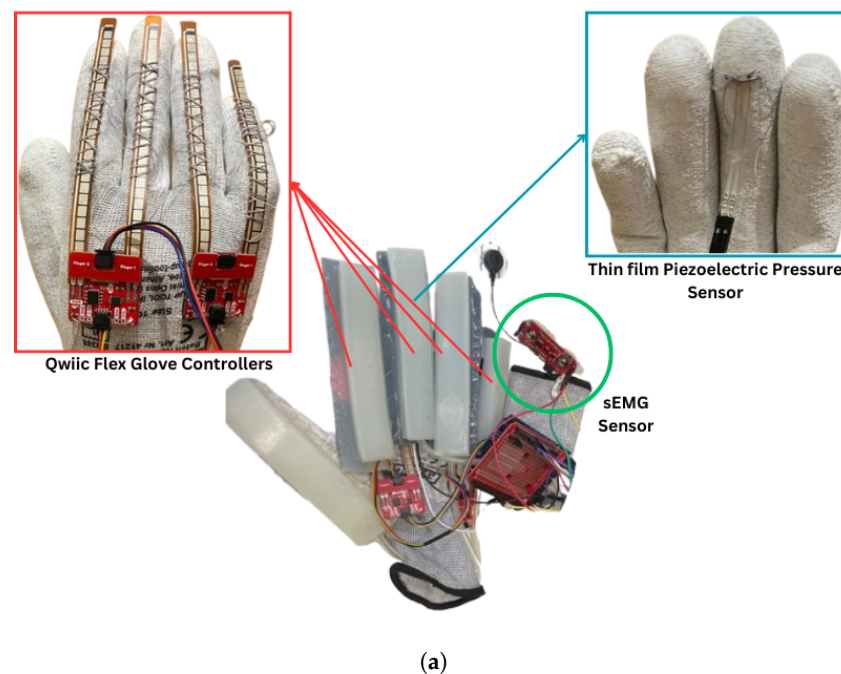


Figure 2. SCI SRG prototype architecture: (a) MSS subsystem and (b) SRPAS subsystem.

2.2.2. SRG Prototype for Post-Stroke Participants

A revised prototype was developed to monitor the kinematics of both hands in post-stroke individuals while also providing targeted pneumatic assistance. These SRGs, integrate ten flex sensors positioned above and along the five fingers of each hand and four thin-film piezoelectric sensors mounted on each thumb and middle finger. The sensors are powered by a 5 V supply, with their output connected to the microcontroller's analog inputs and stabilized by a 10 kOhm pulldown resistor between the ground and the analog input. Additionally, two sEMG Myoware Muscle Sensors are incorporated in each SRG, with each sensor's positive terminal connected to 5 V, the negative terminal to ground, and the signal terminal linked to the corresponding microcontroller analog input. Furthermore, these SRG prototypes include two ADXL345 accelerometers interfaced with an Arduino Mega via the I²C bus (pins 20 and 21). To ensure proper communication, both sensors are wired in parallel on the SDA and SCL lines while sharing the same power and ground connections. Unique I²C addresses were assigned by connecting the AD0 pin of the first sensor to the ground (yielding the default address 0x53) and the AD0 pin of the second sensor to VCC (yielding the address 0x1D).

Complementing the MSS suite, the system also features the SRPAS subsystem, which employs a PneuNet flexible exoskeleton to apply controlled forces to the middle, index, and thumb fingers of each hand. The pneumatic actuation is provided by a single 2.5 lpm air pump that supplies the necessary pressure and a solenoid valve that regulates the airflow for inflation and deflation of the exoskeletons. The valve is controlled by a simple TIP120 transistor circuit—with an accompanying flyback diode for inductive load protection—and its operation is triggered either via the CoMSS or manually via an integrated push button. Power for the actuation components is derived from a regulated 6 V supply, ensuring stable and reliable operation. All sensor and actuation components are integrated into antistatic gloves, as illustrated in Figure 3.

2.3. Experimental Setup and Protocol

Three (3) male chronic SCI patients (ages 21, 62, 53) and three (3) male chronic stroke patients (ages 51, 54, 23) were recruited in the clinical trial protocols [36,37] and, after having obtained informed consent in accordance with the institutionally approved and published study protocol [28], they participated in the validation experiments during March 2024–March 2025. The participants, in addition to basic medical history, were evaluated using an array of clinical measures, which included (a) for SCI patients: the International Standard for Neurological Classification of Spinal Cord Injury (ISNCSCI) [38] and the Spinal Cord Independence Measure (g-SCIM-III) [39,40], and (b) for stroke patients: the Oxford Muscle Scale, the Stroke Impact Scale, version 3.0 (SIS) [41], the Modified Rankin Score-9Q (mRS-9Q) [42], the Timed Up and Go Test at 3 meters (TUG) [43] and the Timed 10-Meter Walk Test [44]. Both groups of patients were administered the Modified Ashworth Scale (MAS) [45] and an array of neuropsychological questionnaires that included the Beck Anxiety Index (BAI) [46], the Beck's Depression Inventory (BDI) [47,48], the Rosenberg Self-Esteem Scale (RSE) [49,50], the Apathy Evaluation Scale (AES) [51], the Vividness of Visual Imagery Questionnaire (VVIQ) [52] and the Kinesthetic and Visual Imagery Questionnaire (KVIQ [53]). The SCI participant evaluation is presented in Table 2 and the post-stroke participant evaluation is presented in Table 3.

Participants reported no prior experience with wearable robotics or with serious game applications. They completed the experimental tasks as described below, taking place in the Thess-AHALL Living Lab [54] of the Medical Physics and Digital Innovation Laboratory. Afterward, they completed an array of questionnaires aiming to evaluate user experience (UX), which included perception of robotics through the GODSPEED Robotics

Questionnaire [55,56], mental effort through the Subjective Mental Effort Questionnaire (SMEQ) [57], and discomfort through the Locally Experienced Discomfort Questionnaire (LED) [58]. Statistical analysis of the answers to the above questionnaires consisted of calculating medians (Mdns) and interquartile ranges (IQRs) for the SCI, stroke, and total participants, which are presented in Table 4 and reported as Mdn(IQR). The number of participants in each leg (3 SCI, 3 stroke, 6 in total) does not allow for further meaningful statistical comparisons, so discussion of those results was based on qualitative observations.

Table 2. Demographic, clinical, and neuropsychological evaluation of the SCI patient participants in the NeuroSuitUp protocol trials.

Participant Info	NSU-01-001	NSU-02-001	NSU-02-002
Age, sex	62, male	21, male	53, male
Time since injury	18 months	24 months	37 years
Injury cause	MVA	MVA	Dive
Complete or incomplete	Complete	Incomplete	Incomplete
NLI	C6	C4	C7
AIS	A	C	B
UEMS	40	22/50	26
LEMS	0	2/50	0
LT	34	41/112	46
PP	32	39/112	84
MAS level UE	1	1+	2
MAS level LE	3	3	0
SCIM total	47/100	41/100	65/100
self-care	10/20	7/20	16/20
respiration and sphincter	26/40	26/40	34/40
mobility	11/40	8/40	15/40
AES	29	27	20
BAI	0	1	1
BDI	7	3	0
RSE	27	24	30
KVIQ	50	49	50
VVIQ	79	80	80

AES: Apathy Evaluation Scale; AIS: ASIA Impairment Scale; BAS: Beck Anxiety Index; BDI: Beck's Depression Index; KVIQ: Kinesthetic and Visual Imagery Questionnaire; LE: Lower Extremity; LEMS: Lower Extremity Motor Score; LT: Light Touch; MAS: Modified Ashworth Scale; MVA: Motor Vehicle Accident; NLI: Neurological Level of Injury; PP: Pin Prick; RSE: Rosenberg Self-Esteem Scale; SCIM: Spinal Cord Independence Measure; UE: Upper Extremity; UEMS: Upper Extremity Motor Subscore; VVIQ: Vividness of Visual Imagery Questionnaire.

Table 3. Demographic, clinical, and neuropsychological evaluation of the stroke patient participants in the HEROES protocol trials.

Participant Info	HER-01-001	HER-02-001	HER-02-002
Age, sex	51, male	54, male	23, male
Time since stroke	31 months	39 months	15 months
Type	Ischemic	Hemorrhagic	Ischemic
Lesion Side	Right	Left	Right
Affected Side	Left	Right	Left
Phenotype	Paraplegia	Paraparesis	Paraparesis
Oxford Arm	0	3+	4
Oxford Hand	0	4+	2–
Oxford Leg	3	4	5–
Oxford Foot	0	3	4
MAS level UE	4	2	3
MAS level LE	2	4	2

Table 3. Cont.

Participant Info	HER-01-001	HER-02-001	HER-02-002
SIS-Strength	6.25	75	75
SIS-Hand	0	70	40
SIS-ADL	37.5	82.5	57.5
SIS-Mobility	77.78	83.33	97.22
SIS-Communication	96.43	100	96.43
SIS-Emotion	58.33	41.67	55.56
SIS-Memory and Thinking	8.16	92.86	85.71
SIS-Participation	37.5	68.75	46.88
SIS-Perceived Recovery	60	60	55
SIS Mean%	42.44	74.90	67.70
mRS-9Q	3	3	2
TUG 3 m (s)	44.74	19.85	11.71
Timed 10 m self (m/s)	0.27	0.63	0.725
Timed 10 m fast (m/s)	0.37	0.94	1.37
AES	43	26	52
BAI	1	2	10
BDI	1	4	25
RSE	14	17	18
KVIQ	13	47	39
KVIQL	8	38	29
KVIQR	10	37	35
VVIQ	40	72	55

ADL: Activities of Daily Living; AES: Apathy Evaluation Scale; BAS: Beck Anxiety Index; BDI: Beck's Depression Index; KVIQ: Kinesthetic and Visual Imagery Questionnaire; KVIQL: KVIQ Left side score; KVIQR: KVIQ Right side score; LE: Lower Extremity; MAS: Modified Ashworth Scale; mRS-9Q: Modified Rankin Score-9Q; Oxford: Oxford scale of muscle strength, lowerst score; RSE: Rosenberg Self-Esteem Scale; SIS: Stroke Impairment Scale; TUG 3 m: Timed Up and Go Test, 3 meters; UE: Upper Extremity; VVIQ: Vividness of Visual Imagery Questionnaire.

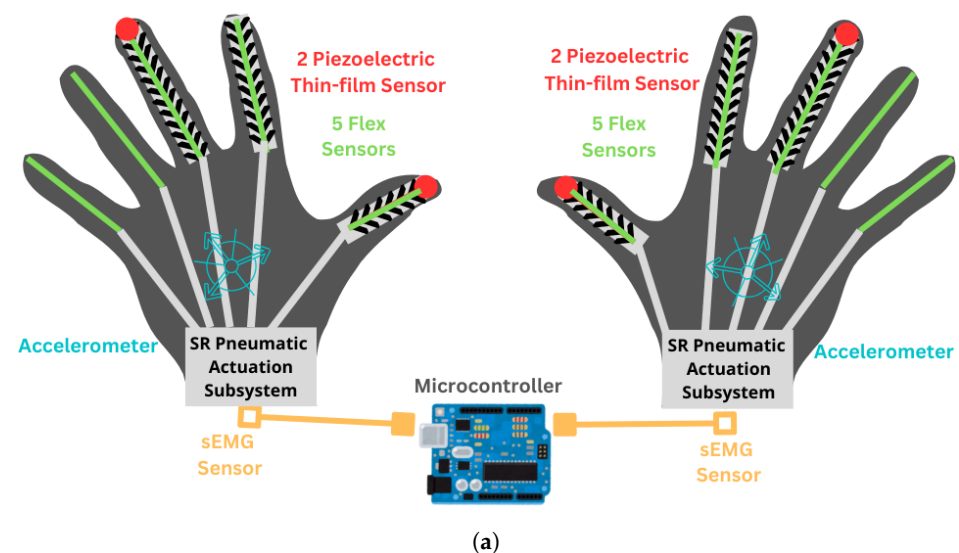


Figure 3. Cont.

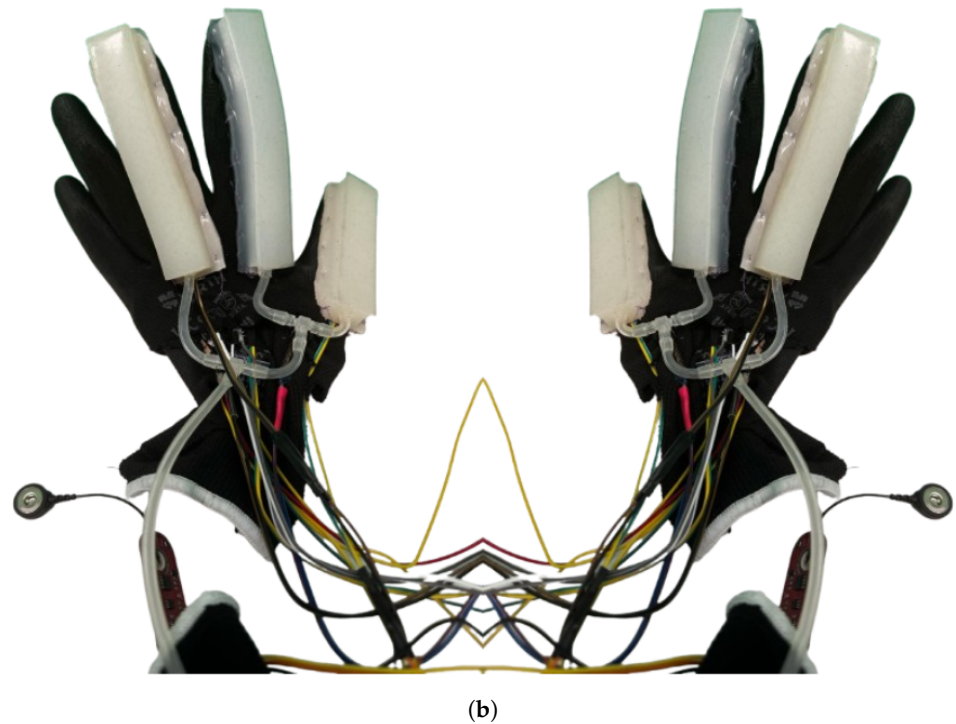


Figure 3. Post-stroke SRGs: (a) architecture and (b) prototype.

Table 4. Perception of robotics, mental effort, discomfort and mental image capacity reported by the patient participants.

Questionnaire	NSU -01-001	NSU -02-001	NSU -02-002	NSU Mdn (IQR)	HER -01-001	HER -02-001	HER -02-002	HER Mdn (IQR)	Total Mdn (IQR)
Godspeed-Total (/120)	73	78	71	73 (3.5)	90	58	70	70 (16)	72 (6.5)
Anthropo-morphism (/25)	12	9	8	9 (2)	14	13	11	13 (1.5)	11.5 (3.25)
Animosity (/30)	14	9	13	13 (2.5)	17	15	10	15 (3.5)	13.5 (4)
Likeability (/25)	17	23	20	20 (3)	20	7	17	17 (6.5)	18.5 (3)
Perceived Intelligence (/25)	15	22	15	15 (3.5)	24	8	17	17 (8)	16 (5.75)
Perceived Safety (/15)	15	15	15	15 (0)	15	15	15	15 (0)	15 (0)
LED	0	0	0	0 (0)	0	0	0	0 (0)	0 (0)
SMEQ	10	0	0	0 (5)	0	10	70	10 (35)	5 (10)
KVIQ	50	49	50	50 (0)	10	37	35	39 (17)	48.5 (9)
VVIQ	79	80	80	80 (0.5)	40	72	55	55 (9)	75.5 (20.5)

IQR: Interquartile Range; KVIQ: Kinesthetic and Visual Imagery Questionnaire; LED: Locally Experienced Discomfort; HER: HEROES; Mdn: Median value; NSU: NeuroSuitUp; SMEQ: Subjective Mental Effort Questionnaire; VVIQ: Vividness of Visual Imagery Questionnaire.

2.4. Data Acquisition and Analysis

Data for the SCI participants group were collected using the ComSS subsystem of the SRG measuring voltage outputs for flexion, pressure, and muscle activity while participants executed 6 repetitions of the Grasping exercises. Each session was timestamped, allowing for the correlation of events with specific time intervals. Categorical data indicating specific events during the exercise, including grasp events (e.g., “Grasp 0”, “Grasp 1”, etc.) and PAS subsystem valve states (e.g., “Valve OPEN”, “Valve CLOSED”) were also annotated. Similarly, data for the post-stroke participants were acquired using the ComSS subsystem integrated into dual SRGs, capturing voltage outputs from the MSS on both hands. Each data file adheres to a standardized format beginning with a timestamp in milliseconds and a phase indicator, followed by sensor readings for each hand. For example, the right-hand data include channels for finger flexion—renamed for clarity as RH_Flex_Thumb, RH_Flex_Index, RH_Flex_Middle, RH_Flex_Ring, and RH_Flex_Little—along with RH_Pressure_Thumb and RH_Pressure_Middle for pressure,

RH_EMG_Raw for muscle activity, and acceleration channels (RH_AccelX, RH_AccelY, RH_AccelZ). The left-hand data mirror this structure with corresponding LH channels. During each session, participants performed three distinct grasping exercises, and the sessions were meticulously timestamped to allow precise correlation between sensor data and specific exercise intervals. In addition, categorical annotations were incorporated to mark key events, such as the initiation and completion of each exercise and the activation of the PAS subsystem, thereby enabling a detailed analysis of motor performance and recovery in post-stroke patients. All recorded data are stored in CSV files, ensuring standardized formatting and ease of access.

2.4.1. Data Preprocessing

The raw data were parsed from a text file and organized into a structured format (DataFrame) using the Python (version 3.10.12) Pandas library (version 2.2.3). Each entry included relevant fields, such as timestamp, event, valve state, sensor voltages, pressure, and EMG readings. The raw time data were converted to numeric values using Python's (version 3.10.12) Pandas library (version 2.2.3). Any rows with missing or non-numeric time values were subsequently removed to ensure a continuous and reliable time base. Data were filtered by each exercise for each condition (non-actuated, right hand actuation, and left hand actuation). In order to facilitate comparison across conditions, the time series were synchronized by aligning the onset of the phase to $t = 0$. Due to potential variability between sensor channels and across participants, each sensor signal was normalized using z-score normalization. Specifically, for each sensor channel, the mean was subtracted and the result was divided by the channel's standard deviation. For channels with zero variance, only the mean was subtracted. This normalization step ensured that differences in scale and sensor drift were minimized, thereby enabling a fair comparison of sensor responses across conditions.

2.4.2. Data Analysis

Data analysis was carried out in Python (version 3.10.12) using Pandas (version 2.2.3) for data handling and Seaborn (version 0.13.2) and Matplotlib (version 3.10.1) for all visualizations. Descriptive statistics—computing mean voltages for each sensor across all grasp events—were used to characterize overall trends and variability in the SCI cohort. Time-series line plots were then produced to depict flex and pressure sensor readings and EMG activity over the course of each exercise; these plots incorporated annotations for individual grasp events and vertical markers indicating valve state transitions, thereby improving interpretability. Finally, a comparative analysis of sensor outputs across successive grasp events was performed to evaluate the SRG's influence on grasp performance and muscle activation.

Variations in sensor outputs across repeated ball-grasping exercises were first examined to evaluate the SRG's effect on grasp performance and muscle activation in the SCI cohort. In the post-stroke group, we extended this analysis by extracting both temporal and spectral features from the recorded signals. Time-domain metrics for the flex sensors included peak amplitude—the maximum voltage attained during each grasp phase—while oscillatory recordings from the sEMG and accelerometer channels were subjected to Fast Fourier Transform to yield frequency-domain descriptors: the dominant frequency (Hz) at which spectral magnitude peaked and the total power (V^2) summed across the spectrum, providing a quantitative measure of overall signal energy.

Principal Component Analysis (PCA) was performed on the flex peak amplitude and sEMG/accelerometer frequency-domain features. This multivariate approach allowed the exploration of the overall sensor response space and the assessment of whether the

frequency profiles of post-stroke participants differed across conditions. PCA facilitated visualization of potential clustering by condition in a reduced dimensional space.

3. Results

The participants' perceptions of the robotics, mental effort, and discomfort are presented in Table 4.

3.1. SCI Participants

Tables 5–7 summarize the mean sensor readings collected during the ball-grasping exercises illustrated in Figures 4–8, as performed by participants wearing the SRG. For each grasp event—labeled sequentially in the “Event” column—the mean values were calculated over a 25s interval while the pneumatic valve remained in the “OPEN” state. Each table reports the following variables: the exact recording time (“Timestamp”), the grasp event identifier (“Event”), and the valve’s operational status (“Valve State”); four digit-specific sensor voltages (“Index,” “Middle,” “Ring,” and “Pinky”); the mean pressure voltage applied by the middle finger (“Pressure”); and the average sEMG amplitude reflecting underlying muscle activity (“EMG”).

These mean values provide critical insights into the performance and functionality of the soft robotic glove during the exercises, allowing for a comprehensive assessment of each finger’s response while the valve was open. By illustrating these averages (Figures 5, 7 and 9), the study aims to evaluate the effectiveness of robotic assistance in enhancing grasping ability among participants.

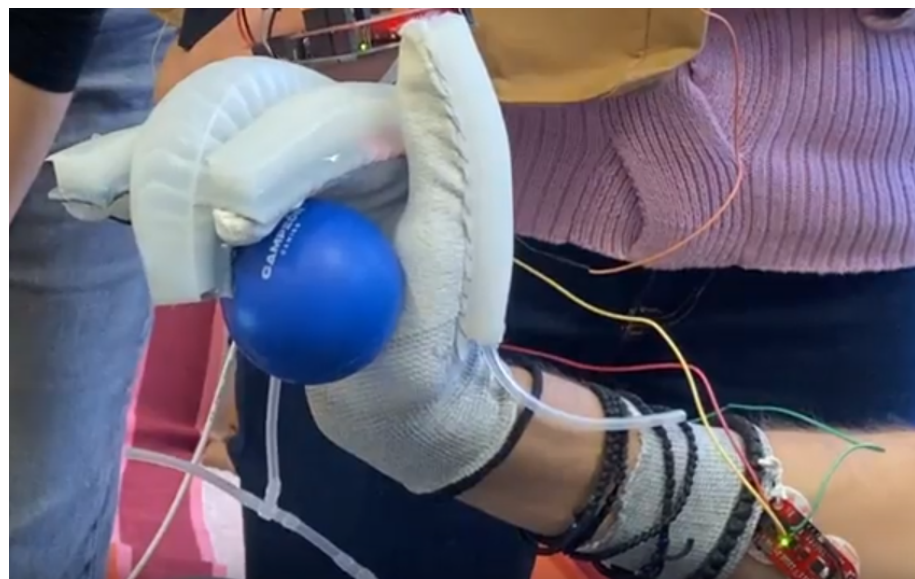


Figure 4. Ball grasping exercise NSU-01-001.

Table 5. Ball grasping exercise—mean NSU-01-001.

Timestamp	Event	Flex Index (V)	Flex Middle (V)	Flex Ring (V)	Flex Little (V)	Pressure Middle (V)	EMG (V)
11:38:47	0	4.17	4.04	3.96	4.31	0.01	0.52
11:39:12	1	4.16	4.09	4.04	4.27	0.01	0.75
11:39:37	2	4.21	4.26	3.82	4.22	0.01	0.48
11:40:02	3	4.18	4.13	4.03	4.30	0.01	0.40
11:40:27	4	4.14	4.01	3.93	4.32	0.01	0.55
11:40:52	5	4.18	4.25	3.91	4.30	0.01	0.66

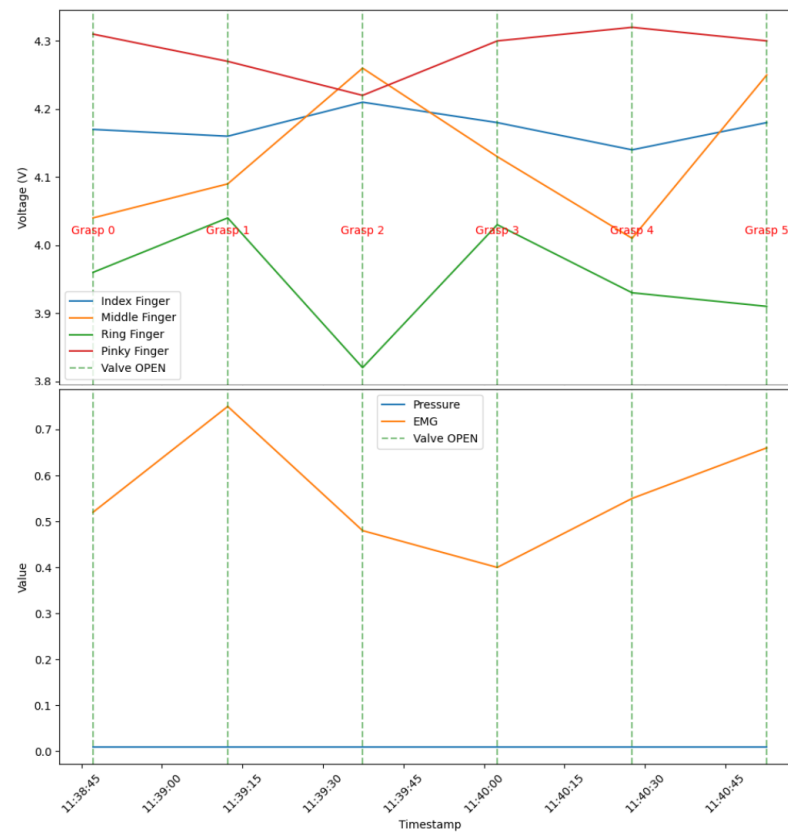


Figure 5. Ball grasping exercise—mean NSU-01-001.



Figure 6. Ball grasping exercise NSU-02-001.

Table 6. Ball grasping exercise—mean NSU-02-001.

Timestamp	Event	Flex Index (V)	Flex Middle (V)	Flex Ring (V)	Flex Pinky (V)	Pressure (V)	EMG (V)
10:23:00	0	4.32	4.40	4.21	4.36	0.02	0.37
10:23:25	1	4.35	4.29	4.25	4.25	0.01	0.66
10:23:50	2	4.41	4.36	4.30	4.30	0.01	0.79
10:24:15	3	4.35	4.35	4.22	4.36	0.02	0.43
10:24:40	4	4.36	4.41	4.20	4.33	0.02	0.43
10:25:05	5	4.28	4.18	3.72	3.74	0.01	0.63

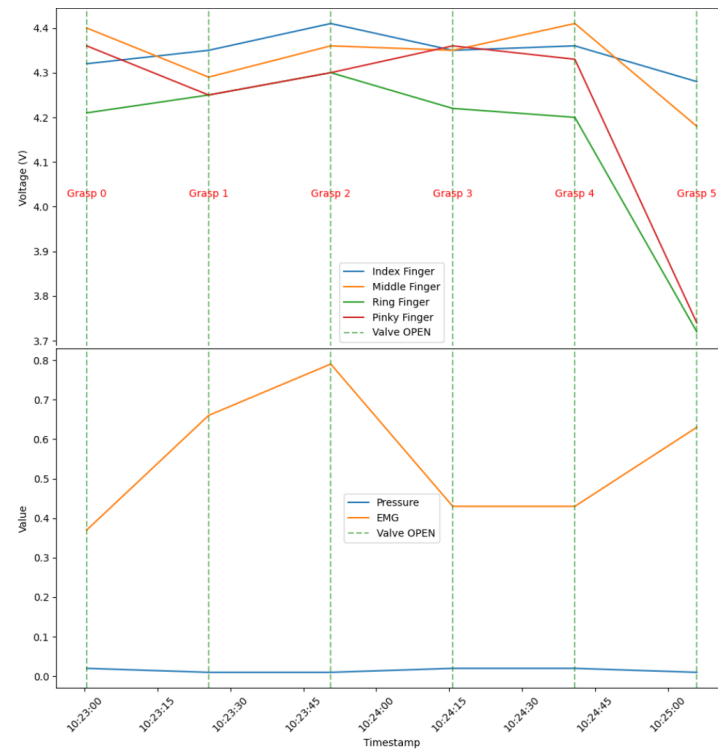


Figure 7. Ball grasping exercise—mean NSU-02-001.

Table 7. Ball grasping exercise—mean NSU-02-002.

Timestamp	Event	Flex Index (V)	Flex Middle (V)	Flex Ring (V)	Flex Pinky (V)	Pressure (V)	EMG (V)
16:52:12	0	4.14	4.25	4.06	4.26	0.03	0.43
16:52:37	1	4.27	4.33	4.14	4.27	3.59	4.79
16:53:02	2	4.12	4.31	4.17	4.28	3.70	4.57
16:53:27	3	4.13	4.17	4.06	4.26	0.02	0.83
16:53:52	4	4.10	4.15	4.03	4.24	0.02	0.53
16:54:17	5	4.21	4.27	4.10	4.19	0.02	0.46

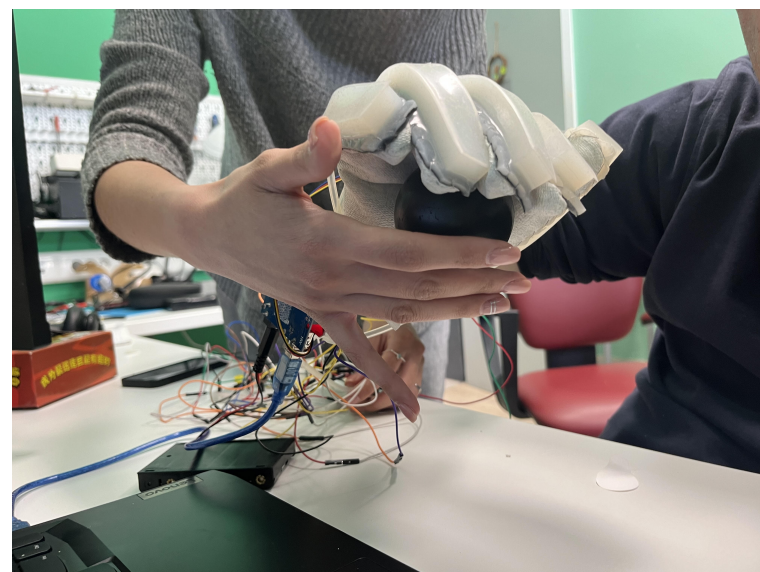


Figure 8. Ball grasping exercise NSU-02-002.

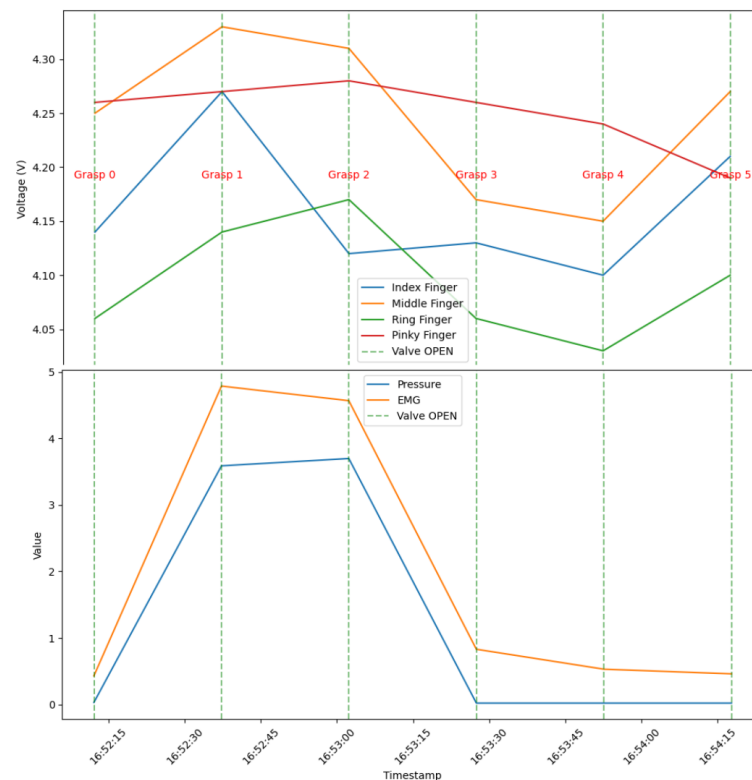


Figure 9. Ball grasping exercise—mean NSU-02-002.

3.2. Post-Stroke Participants

In the post-stroke cohort, sensor data were analyzed across three conditions—non-actuated, left-actuated, and right-actuated—to evaluate how the SRPAS assistance modulates hand function during the three grasping tasks: Ball grasping (Figures 10–14; Tables 8–10), glass grip (Figures 15–19; Tables 11–13), paper grip (Figures 20–25; Tables 14–16). Figures of flex sensor peak amplitude for all participants and exercises (Figures 10, 15 and 21) illustrate distinct patterns of grip strength, while FFT spectral plots (Figures 11–13, 16–18, 22–24) reveal differences in the frequency content of the signals, indicating shifts in dominant frequencies and power distributions under actuated conditions. Additionally, for each exercise time and frequency domain features were calculated (Tables 8–16) and PCA plots (Figures 14, 19 and 25) performed demonstrating that the overall sensor response space separates the conditions, further supporting the observation that actuation alters neuromuscular activation. Collectively, these results underscore the potential of the SRG system to improve motor performance in post-stroke individuals by modulating both the temporal and spectral characteristics of the collected MSS signals.

3.2.1. Ball Grasping

The ball grasping exercise was designed to assess the participant's ability to generate and modulate grip force in a simple, spherical object manipulation. During each trial, subjects reached for and closed their hand around a standardized 5-cm diameter soft ball, allowing measurement of flex-sensor peak amplitude as an index of maximal voluntary contraction. By comparing non-actuated, left-actuated, and right-actuated conditions, we quantified how the SRPAS assistance affects the grasping generation. Time-domain features and frequency-domain characteristics were extracted from each trial and subsequently projected into PCA plots to visualize shifts in the sensor response under different actuation states. This introductory analysis lays the groundwork for understanding the neuromuscular adaptations induced by SRG assistance in post-stroke hand function rehabilitation.

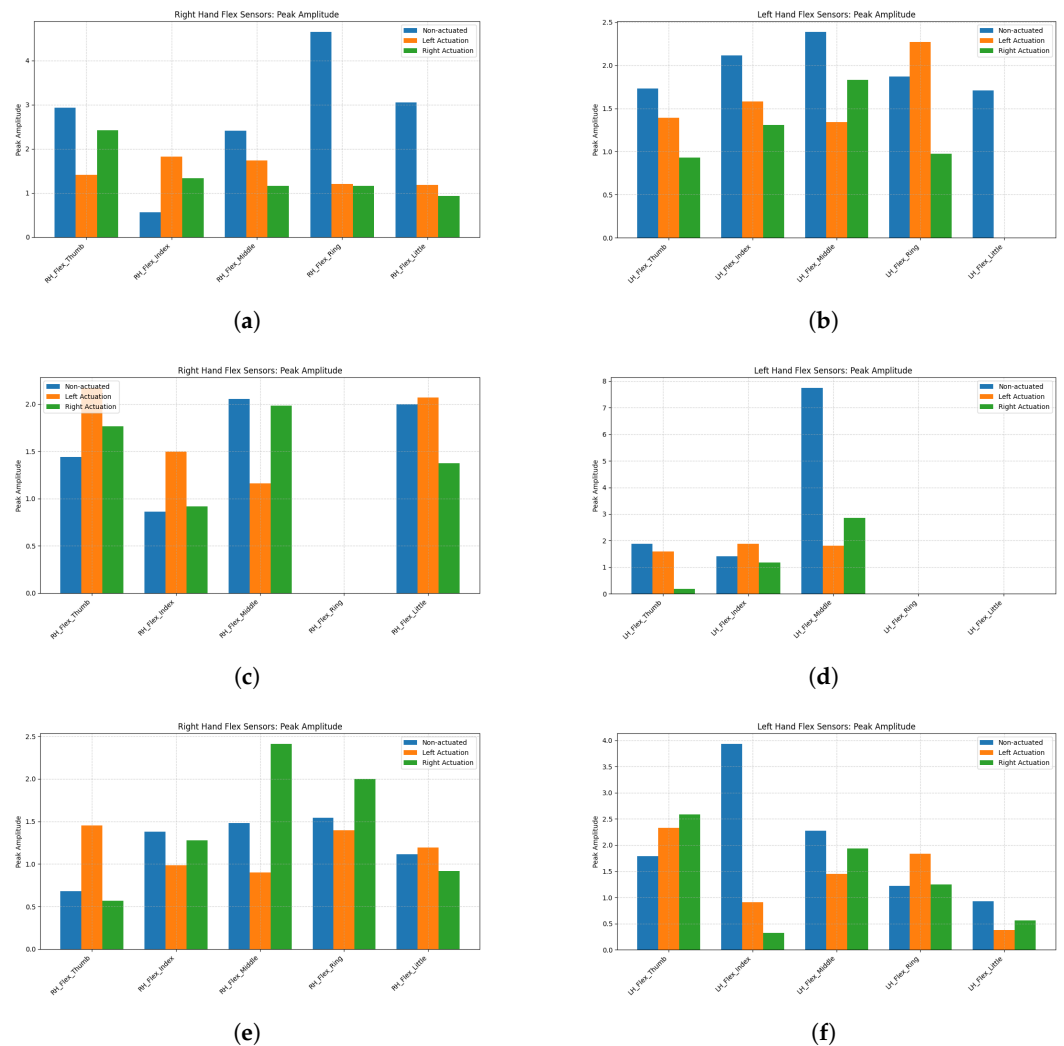


Figure 10. Flex peak amplitude—ball grasping: (a) HER-01-001 right hand; (b) HER-01-001 left hand; (c) HER-02-001 right hand; (d) HER-02-001 left hand; (e) HER-02-002 right hand; (f) HER-02-002 left hand.

Table 8. Time-domain peak amplitude features for flex and pressure sensors—ball grasping exercise.

Participant	Condition	LH Flex PeakAmp (V)	RH Flex PeakAmp (V)
HER-01-001	Non-actuated	1.962385	2.720541
HER-01-001	Left Actuation	1.315986	1.473494
HER-01-001	Right Actuation	1.009055	1.402194
HER-02-001	Non-actuated	2.209620	1.271211
HER-02-001	Left Actuation	1.057016	1.379960
HER-02-001	Right Actuation	0.844384	1.207886
HER-02-002	Non-actuated	2.028898	1.241410
HER-02-002	Left Actuation	1.380525	1.186795
HER-02-002	Right Actuation	1.331171	1.435131

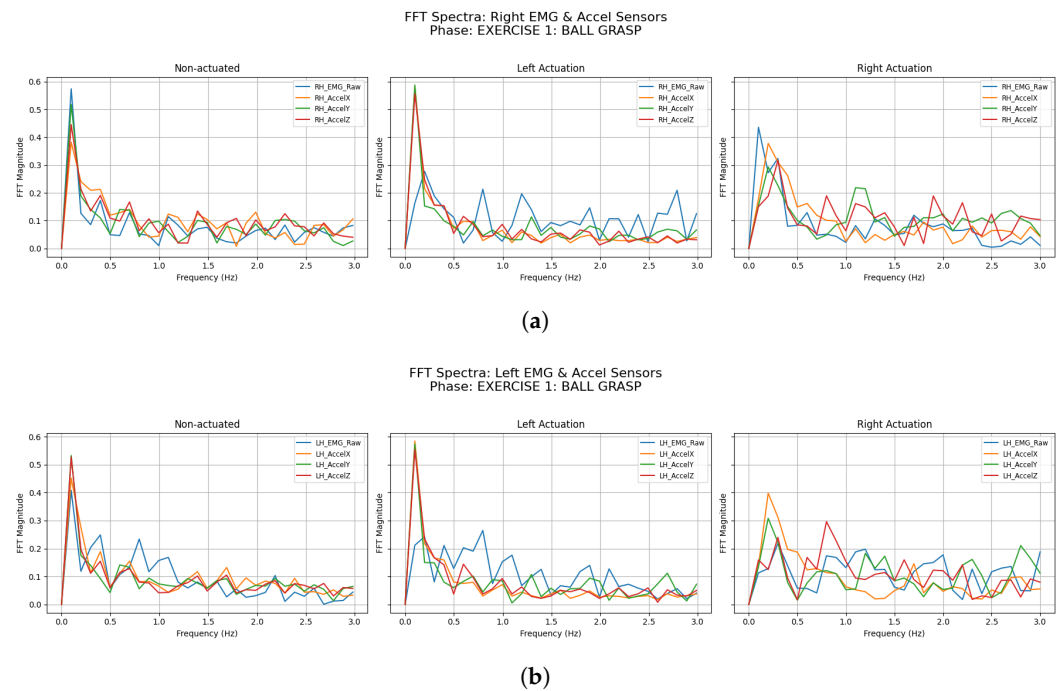


Figure 11. HER-01-001—EMG and accelerometer data FFT spectra—ball grasping: (a) right hand and (b) left hand.

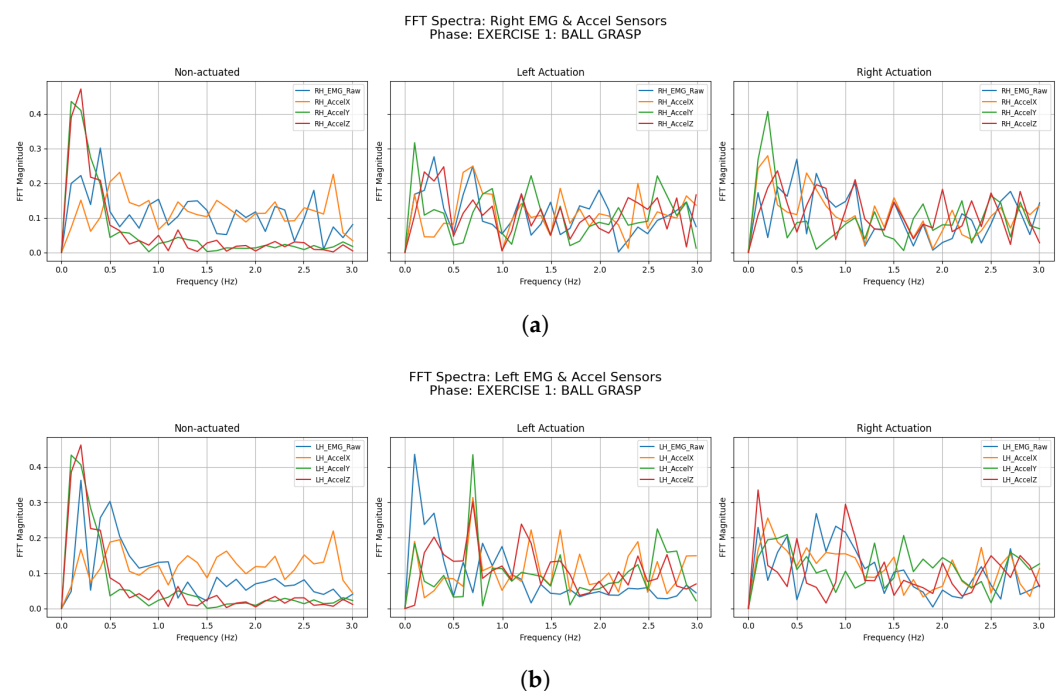


Figure 12. HER-02-001—EMG and accelerometer data FFT spectra—ball grasping: (a) right hand and (b) left hand.

3.2.2. Glass Grip

The glass grip exercise measured how well participants flex their fingers around a cylindrical object resembling a drinking glass. Participants reached out and wrapped their fingers around a 6 cm-diameter, 10 cm-tall cardboard tube engineered to simulate the feel and weight of a filled glass. We compared trials without actuation to those with left-side and right-side assistance from the SRPAS to see if and how the device alters grip coordination. From each trial, we extracted both time-domain and frequency-domain features, then used PCA to reveal clear groupings of sensor patterns under each condition.

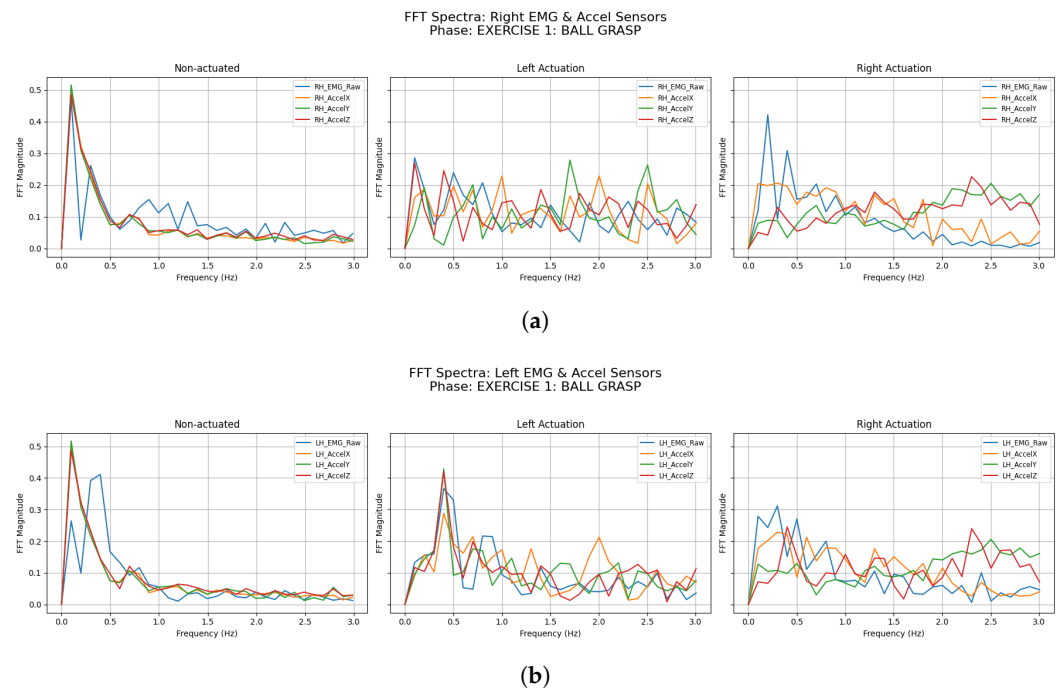


Figure 13. HER-02-002—EMG and accelerometer data FFT spectra—ball grasping: (a) right hand and (b) left hand.

Table 9. Frequency-domain features for accelerometer sensors—ball grasping exercise.

Participant	Condition	LH Accel Dominant Freq (Hz)	LH Accel Total Power (V^2)	RH Accel Dominant Freq (Hz)	RH Accel Total Power (V^2)
HER-01-001	Non-actuated	0.099354	1830.000000	0.099354	1830.000000
HER-01-001	Left Actuation	0.099562	1888.322165	0.099562	1888.413307
HER-01-001	Right Actuation	0.399840	1830.000000	0.233240	1830.000000
HER-02-001	Non-actuated	1.035197	1889.605088	0.300541	1890.473058
HER-02-001	Left Actuation	0.696933	1880.280903	0.398248	1884.340688
HER-02-001	Right Actuation	0.233754	1875.938184	0.233754	1881.659661
HER-02-002	Non-actuated	0.099562	1889.261171	0.099562	1888.783603
HER-02-002	Left Actuation	0.400721	1889.190340	1.268951	1885.319101
HER-02-002	Right Actuation	1.068590	1867.917990	1.703066	1871.429125

Table 10. Frequency-domain features for EMG sensors—ball grasping exercise.

Participant	Condition	LH EMG Dominant Freq (Hz)	LH EMG Total Power (V^2)	RH EMG Dominant Freq (Hz)	RH EMG Total Power (V^2)
HER-01-001	Non-actuated	0.099354	1830.000000	0.099354	1830.000000
HER-01-001	Left Actuation	0.796495	1890.282550	0.199124	1854.152767
HER-01-001	Right Actuation	0.299880	1830.000000	0.099960	1830.000000
HER-02-001	Non-actuated	0.200361	1886.492856	0.400721	1889.190431
HER-02-001	Left Actuation	0.099562	1887.675208	0.298686	1883.218107
HER-02-001	Right Actuation	0.701262	1890.172429	0.500902	1888.627353
HER-02-002	Non-actuated	0.398248	1890.866749	0.099562	1890.583968
HER-02-002	Left Actuation	0.400721	1890.911656	0.100180	1823.081392
HER-02-002	Right Actuation	0.300541	1886.715516	0.200361	1890.946518

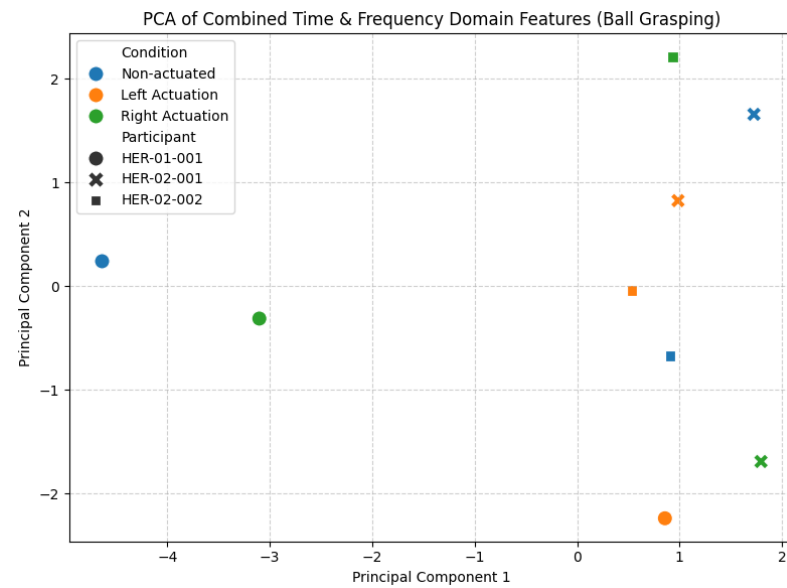


Figure 14. Ball grasping exercise—PCA time and frequency domain features.

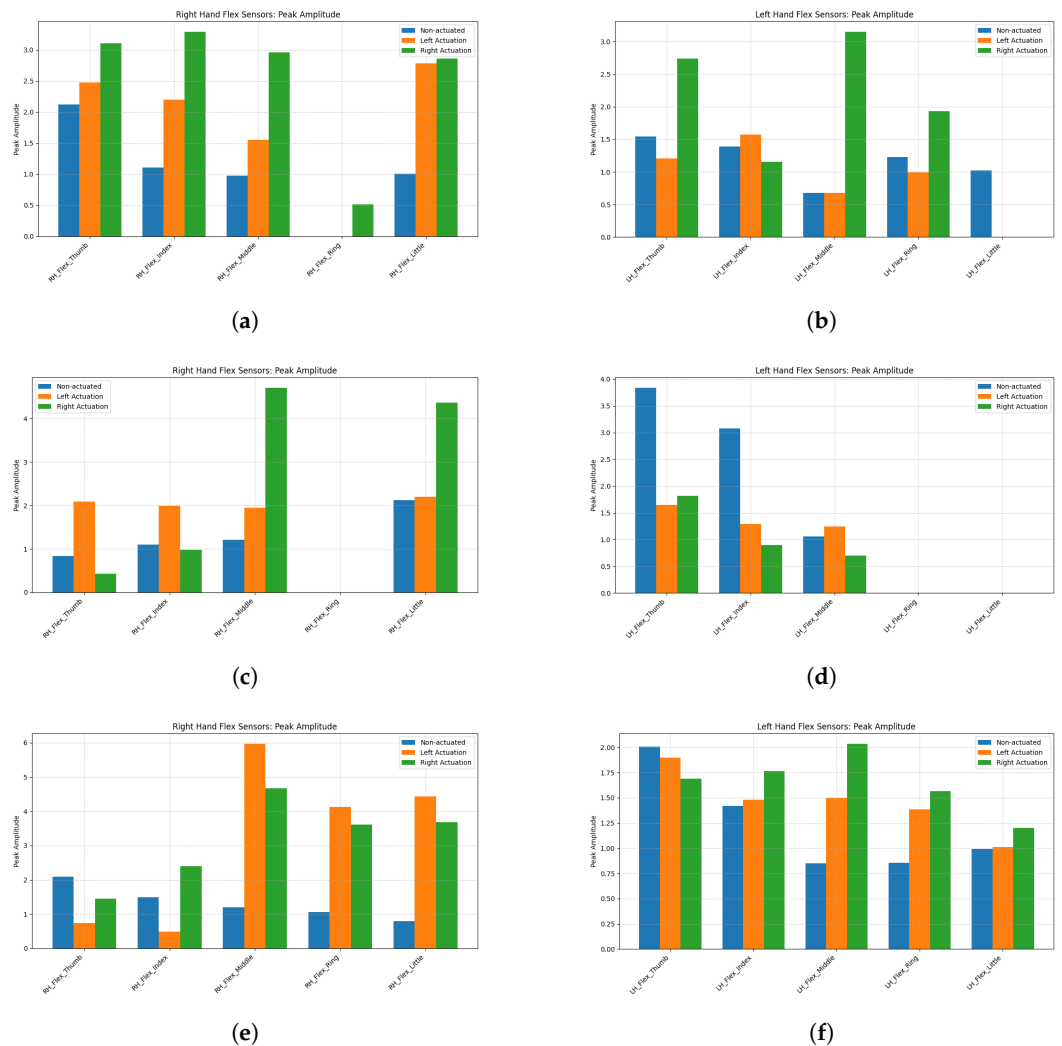


Figure 15. Flex peak amplitude—glass grip: (a) HER-01-001 right hand; (b) HER-01-001 left hand; (c) HER-02-001 right hand; (d) HER-02-001 left hand; (e) HER-02-002 right hand; (f) HER-02-002 left hand.

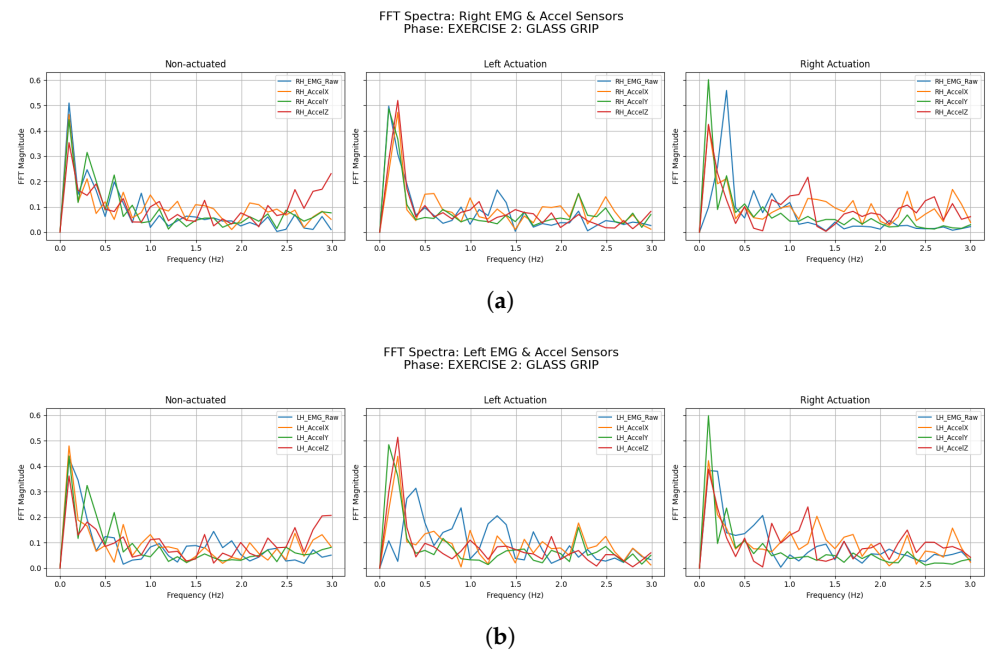


Figure 16. HER-01-001—EMG and accelerometer data FFT spectra—glass grip: (a) right hand and (b) left hand.

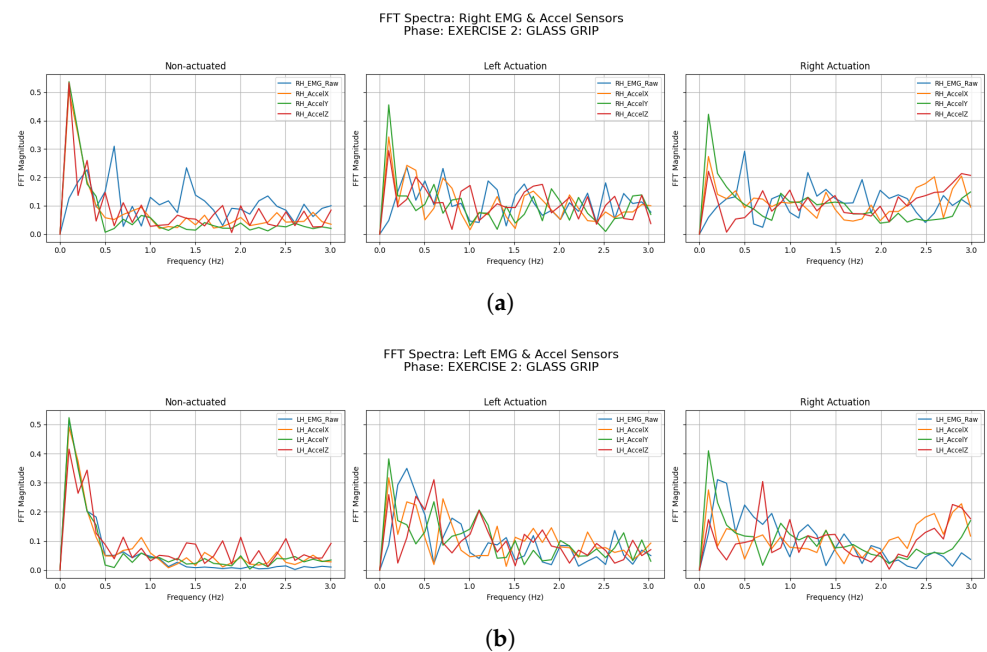


Figure 17. HER-02-001—EMG and accelerometer data FFT spectra—glass grip: (a) right hand and (b) left hand.

Table 11. Time-domain peak amplitude feature for flex sensors—glass grip.

Participant	Condition	LH Flex PeakAmp (V)	RH Flex PeakAmp (V)
HER-01-001	Non-actuated	1.171659	1.042512
HER-01-001	Left Actuation	0.889562	1.801664
HER-01-001	Right Actuation	1.796198	2.546574
HER-02-001	Non-actuated	1.593946	1.052788
HER-02-001	Left Actuation	0.835917	1.644414
HER-02-001	Right Actuation	0.682261	2.097046
HER-02-002	Non-actuated	1.223903	1.329954
HER-02-002	Left Actuation	1.455255	3.152663
HER-02-002	Right Actuation	1.651295	3.163606

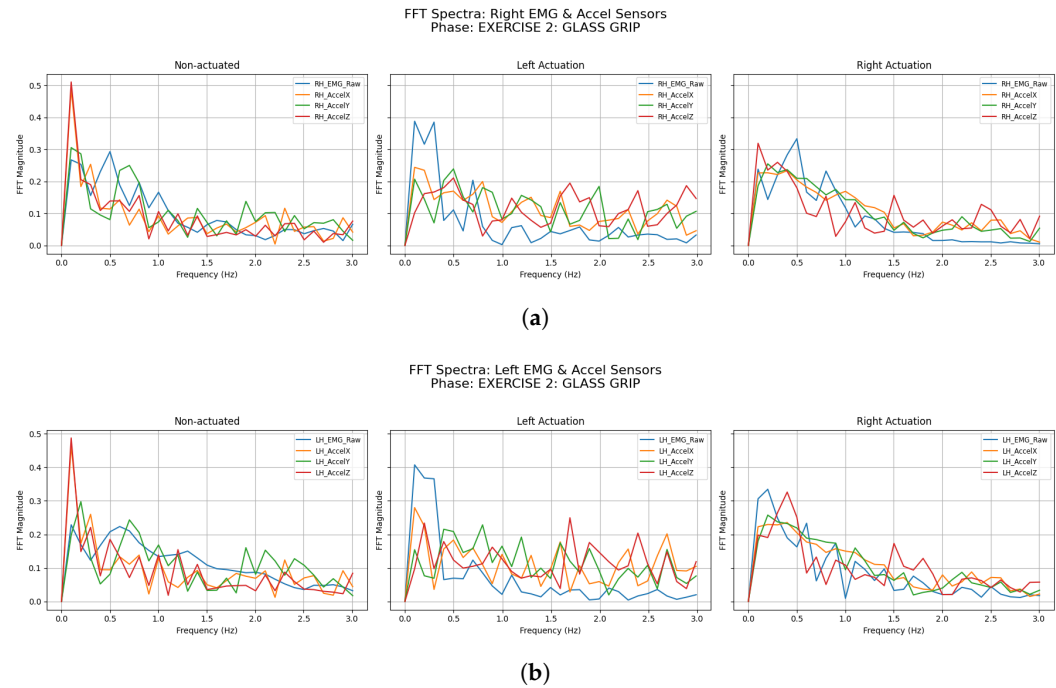


Figure 18. HER-02-002—EMG and accelerometer data FFT spectra—glass grip: (a) right hand and (b) left hand.

Table 12. Frequency-domain features for accelerometer sensors—glass grip.

Participant	Condition	LH Accel Dominant Freq (Hz)	LH Accel Total Power (V ²)	RH Accel Dominant Freq (Hz)	RH Accel Total Power (V ²)
HER-01-001	Non-actuated	0.099562	1878.102515	0.099562	1888.174548
HER-01-001	Left Actuation	0.165937	1890.988507	0.099562	1887.806243
HER-01-001	Right Actuation	0.099960	1830.000000	0.099960	1830.000000
HER-02-001	Non-actuated	0.100180	1890.451842	0.100180	1890.495245
HER-02-001	Left Actuation	0.268817	1889.718931	0.100806	1874.682196
HER-02-001	Right Actuation	0.298686	1890.608011	0.099562	1832.358876
HER-02-002	Non-actuated	0.133574	1889.471475	0.100180	1890.800619
HER-02-002	Left Actuation	0.862870	1890.057772	0.365060	1878.191281
HER-02-002	Right Actuation	0.333934	1890.839838	0.233754	1889.583496

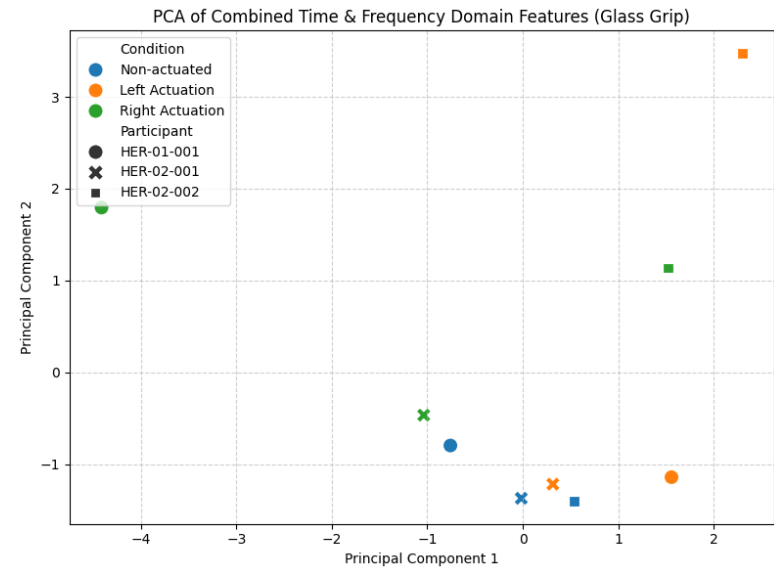


Figure 19. Glass grip exercise—PCA time and frequency domain features.

3.2.3. Paper Grip

In the paper grip exercise, participants used their fingers as visualized in Figure 20 to lift and hold two cardboard papers. We ran this under three conditions—no actuation, left-side SRPAS assistance, and right-side assistance—to explore how the device influences the performance of this grip. From each trial, we extracted time-domain measures and frequency-domain descriptors. A PCA of these features then revealed clear grouping of sensor responses by condition, highlighting how actuation assists the hand coordination in this grip.



Figure 20. Paper grip—exercise example.

Table 13. Frequency-domain features for EMG sensors—glass grip.

Participant	Condition	LH EMG Dominant Freq (Hz)	LH EMG Total Power (V^2)	RH EMG Dominant Freq (Hz)	RH EMG Total Power (V^2)
HER-01-001	Non-actuated	0.099562	1878.102515	0.099562	1888.174548
HER-01-001	Left Actuation	0.398248	1890.988507	0.099562	1887.806243
HER-01-001	Right Actuation	0.099960	1830.000000	0.299880	1830.000000
HER-02-001	Non-actuated	0.100180	1890.451842	0.400721	1890.495245
HER-02-001	Left Actuation	0.302419	1889.718931	0.302419	1862.682196
HER-02-001	Right Actuation	0.199124	1890.608011	0.497810	1887.605027
HER-02-002	Non-actuated	0.100180	1889.471475	0.500902	1890.800619
HER-02-002	Left Actuation	0.099562	1890.057772	0.099562	1890.756482
HER-02-002	Right Actuation	0.200361	1890.839838	0.500902	1890.997388

Table 14. Flex sensor time-domain features for flex sensors—paper grip.

Participant	Condition	LH Flex PeakAmp (V)	RH Flex PeakAmp (V)
HER-01-001	Non-actuated	1.962385	2.720541
HER-01-001	Left Actuation	1.315986	1.473494
HER-01-001	Right Actuation	1.009055	1.402194
HER-02-001	Non-actuated	2.209620	1.271211
HER-02-001	Left Actuation	1.057016	1.379960
HER-02-001	Right Actuation	0.844384	1.207886
HER-02-002	Non-actuated	2.028898	1.241410
HER-02-002	Left Actuation	1.380525	1.186795
HER-02-002	Right Actuation	1.331171	1.435131

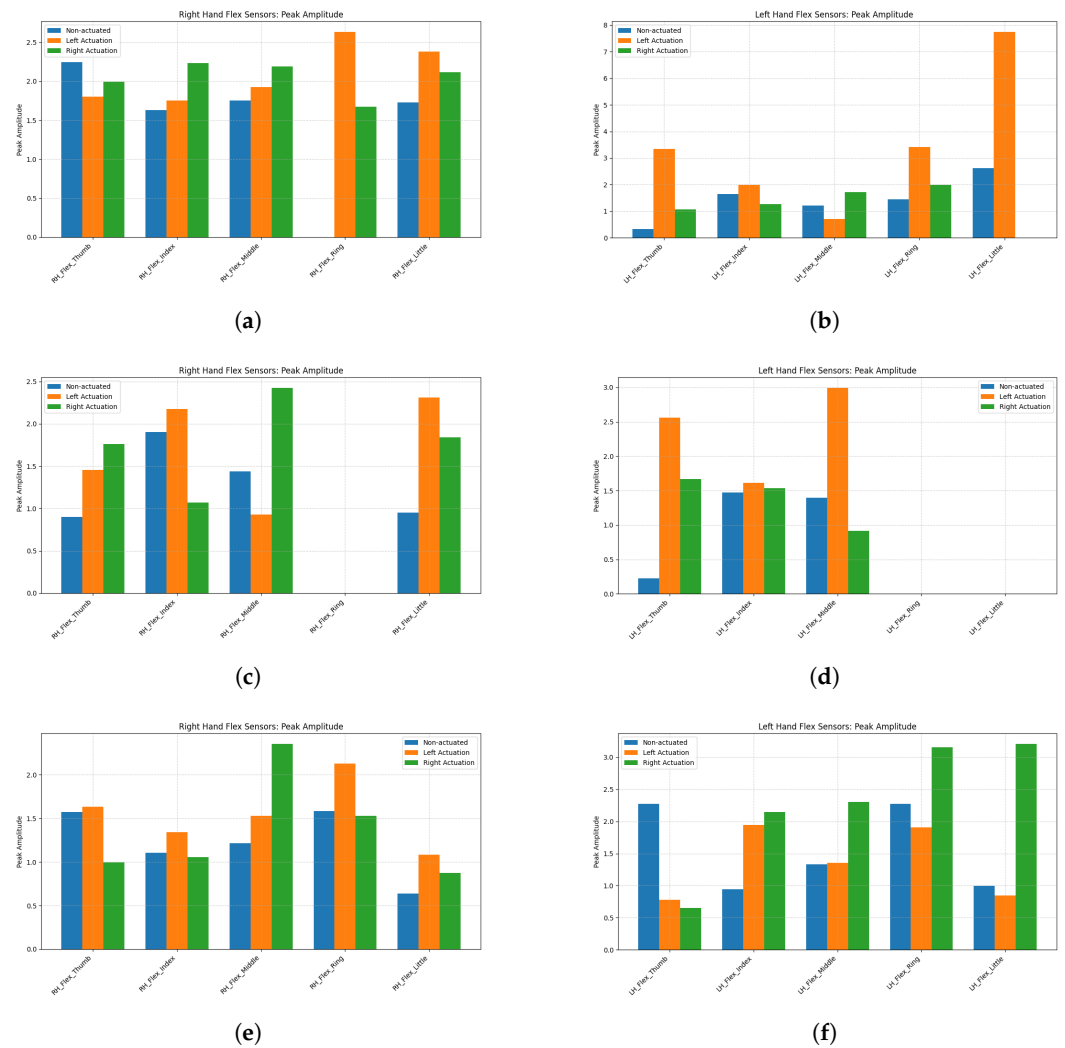


Figure 21. Flex peak amplitude—paper grip: (a) HER-01-001 right hand; (b) HER-01-001 left hand; (c) HER-02-001 right hand; (d) HER-02-001 left hand; (e) HER-02-002 right hand; (f) HER-02-002 left hand.

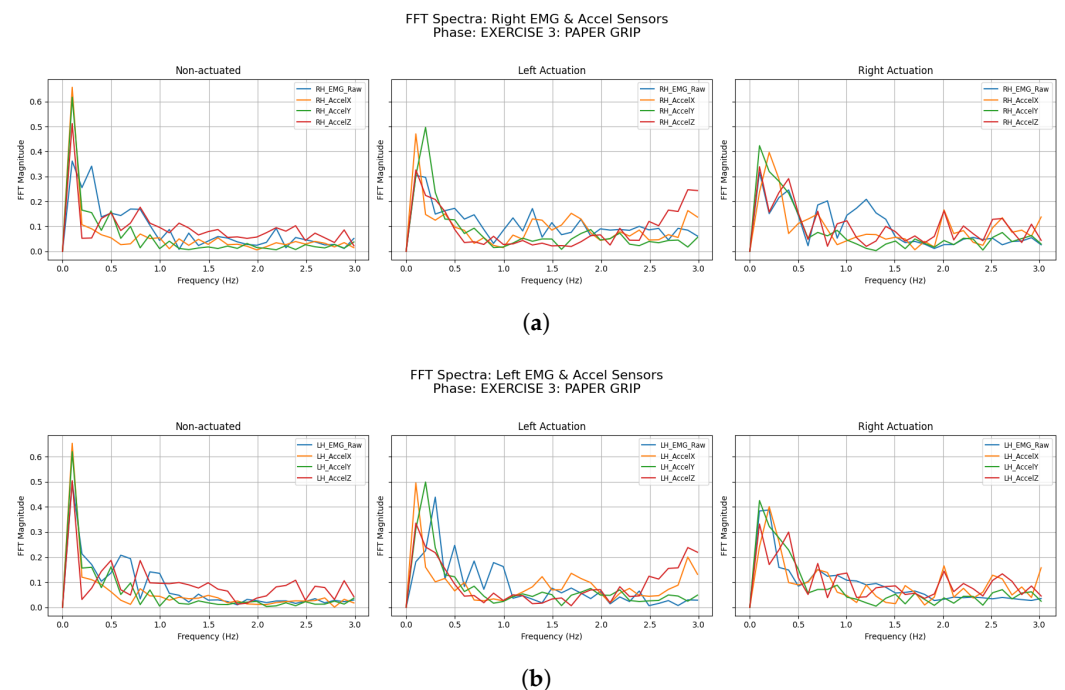


Figure 22. HER-01-001—EMG and accelerometer data FFT spectra—paper grip: (a) right hand and (b) left hand.

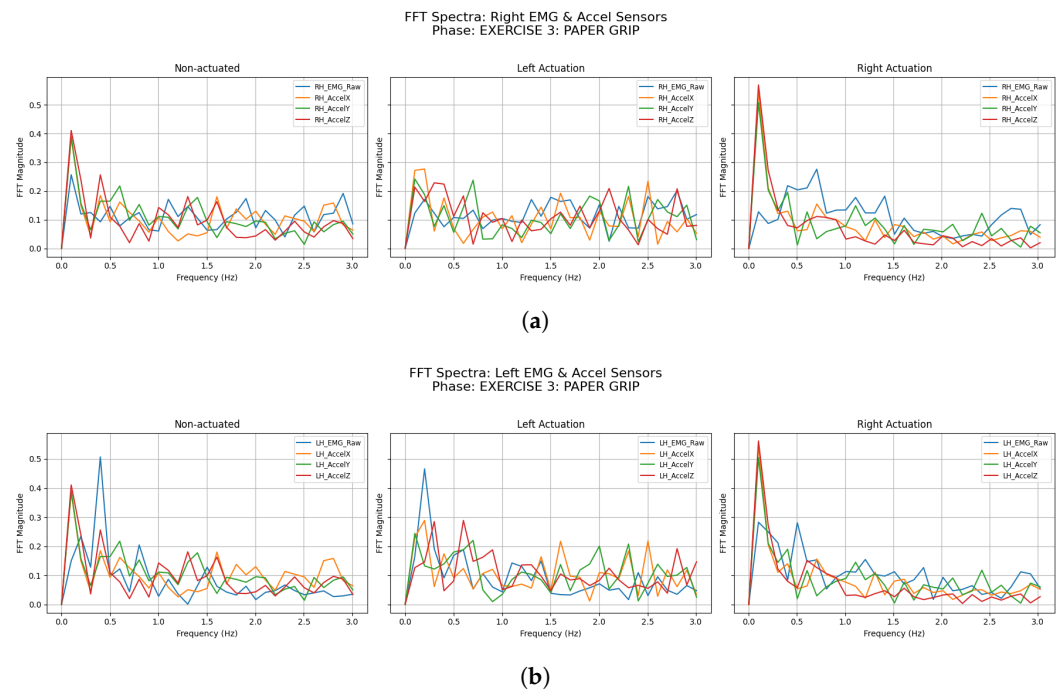


Figure 23. HER-02-001—EMG and accelerometer data FFT spectra—paper grip: (a) right hand and (b) left hand.

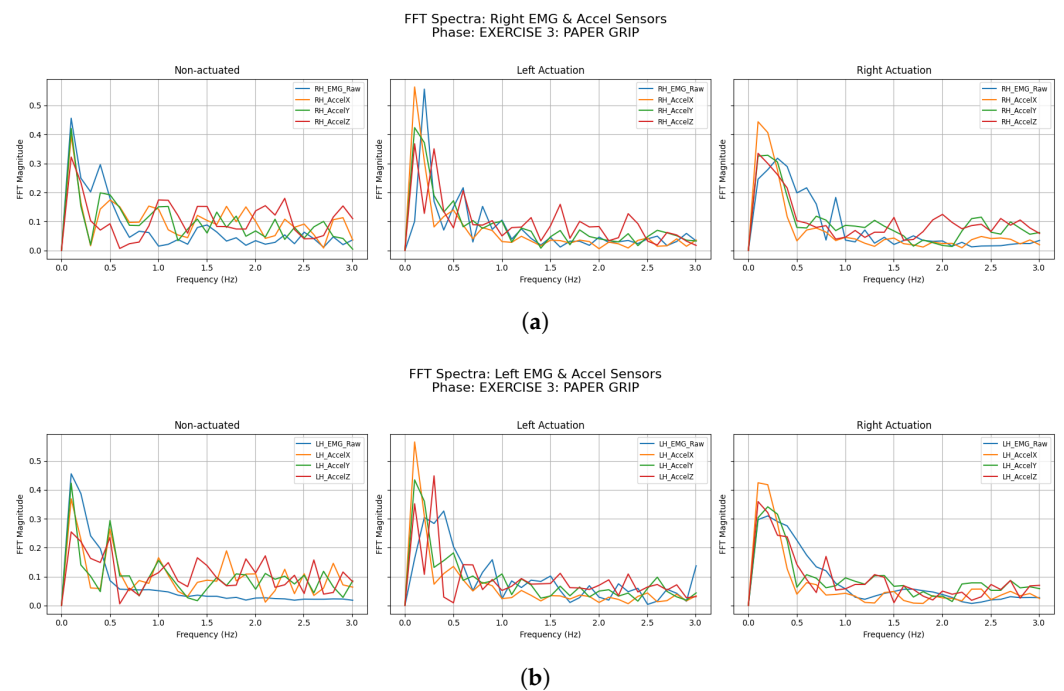


Figure 24. HER-02-002—EMG and accelerometer data FFT spectra—paper grip: (a) right hand and (b) left hand.

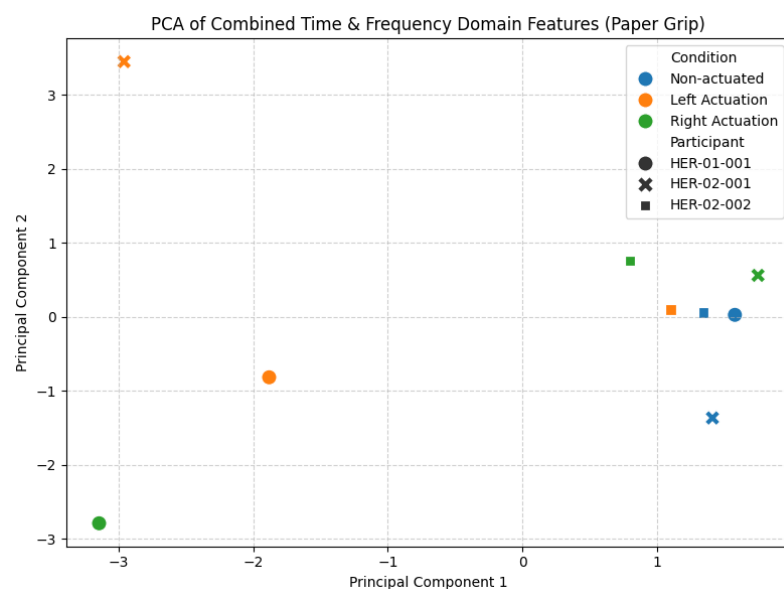
Finally, to explore the monotonic relationships between participants' subjective ratings and their biomechanical signals, we computed Spearman rank-order correlations across all six participants, the the results of which are presented in Figure 26 below.

Table 15. Frequency-domain features for accelerometer sensors—paper grip.

Participant	Condition	LH Accel Dominant Freq (Hz)	LH Accel Total Power (V^2)	RH Accel Dominant Freq (Hz)	RH Accel Total Power (V^2)
HER-01-001	Non-actuated	0.099354	1830.000000	0.099354	1830.000000
HER-01-001	Left Actuation	0.099562	1888.322165	0.099562	1888.413307
HER-01-001	Right Actuation	0.399840	1830.000000	0.233240	1830.000000
HER-02-001	Non-actuated	1.035197	1889.605088	0.300541	1890.473058
HER-02-001	Left Actuation	0.696933	1880.280903	0.398248	1884.340688
HER-02-001	Right Actuation	0.233754	1875.938184	0.233754	1881.659661
HER-02-002	Non-actuated	0.099562	1889.261171	0.099562	1888.783603
HER-02-002	Left Actuation	0.400721	1890.911656	0.100180	1823.081392
HER-02-002	Right Actuation	1.068590	1867.917990	1.703066	1871.429125

Table 16. Frequency-domain features for EMG sensors—paper grip.

Participant	Condition	LH EMG Dominant Freq (Hz)	LH EMG Total Power (V^2)	RH EMG Dominant Freq (Hz)	RH EMG Total Power (V^2)
HER-01-001	Non-actuated	0.099354	1830.000000	0.099354	1830.000000
HER-01-001	Left Actuation	0.796495	1890.282550	0.199124	1854.152767
HER-01-001	Right Actuation	0.299880	1830.000000	0.099960	1830.000000
HER-02-001	Non-actuated	0.200361	1886.492856	0.400721	1889.190431
HER-02-001	Left Actuation	0.099562	1887.675208	0.298686	1883.218107
HER-02-001	Right Actuation	0.701262	1890.172429	0.500902	1888.627353
HER-02-002	Non-actuated	0.398248	1890.866749	0.099562	1890.583968
HER-02-002	Left Actuation	0.400721	1890.911656	0.100180	1823.081392
HER-02-002	Right Actuation	0.300541	1886.715516	0.200361	1890.946518

**Figure 25.** Paper grip exercise—PCA time and frequency domain features.

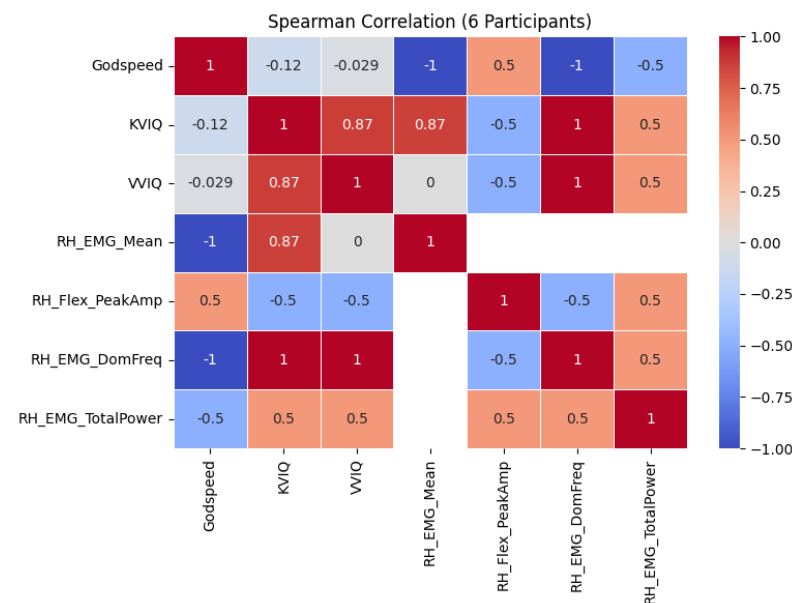


Figure 26. Spearman correlation of participant data.

4. Discussion

The findings presented in this manuscript provide a thorough investigation of grasping task performance among SCI and stroke patients, with the SRPAS that we developed assisting with each task. All participants in our study had established chronic impaired function of the hands (at least 18 months old) and plateaued during their rehabilitation process. SCI participants had impaired function of both hands with UEMS ranging from 22/50 to 40/50 and one among them demonstrated a complete injury. Similarly, stroke participants had their affected hand's muscle function evaluated with the Oxford scale from 0 to 4+, with one participant being left-side plegic. Due to their lesions, the patients demonstrated varied levels of spasticity of the affected hands, ranging from 1 to 4 in the MAS, further impairing their achievable ROM during hand-related activities and SRPAS-assisted tasks.

The tasks involved in our investigation were designed to resemble ADLs [6,29,59,60], such as repeated and intermittent performing of spherical, cylindrical, and lumbrical gripping, using objects such as balls, glass and paper. Our SCI participants' disability was documented through the SCIM questionnaire, where their scores in the self-care category indicated severe to moderate disability [61,62], ranging from 7 to 16 out of 20. Similarly, the disability of our stroke participants was validated through SIS [63,64], where their cores in the hand domain ranged from 0% to 70% and their scores in the ADL domain ranged from 37.5% to 57.5%. As a result of their impaired motor function, these individuals would not have been able to complete the activities using their affected hands, without the mechanical assistance provided by the SRPAS. The finger flexion values that we acquired for the index, middle, ring, and pinky fingers demonstrate a stable and consistent grip performance, which was made feasible using pneumatic actuation. The relatively uniform flexion across different fingers emphasizes the necessity of this external mechanical help in allowing coordinated finger motions in order to compensate for the lack of individual motor control and reduced ROM.

The piezoelectric pressure and sEMG measurements demonstrate the relevance of pneumatic actuation in grasping performance. The increased pressure values recorded, notably during Events 1 and 2, suggest that the SRPAS was actively aiding the SCI participants' grasp. This increased pressure shows that the actuation delivered the necessary mechanical force, allowing participants to execute a task that would otherwise be impossible for them to achieve unaided. The sEMG data, which correlate to muscle activation, demonstrate

that, while the patients exerted some muscular activity, the robotic aid took on a significant portion of the effort, as reflected in the low-to-moderate sEMG values in relation to the task's mechanical demands.

Figures 14–25 present PCA projections of combined time-domain (flex peak amplitude) and frequency-domain (sEMG/accelerometer dominant frequency, total power) features for each exercise and actuation condition. In all plots, PC1 and PC2 capture the largest and second-largest portions of variance, respectively. Color coding denotes non-actuated, left-actuated, and right-actuated trials, while marker shapes distinguish participants. In the ball-grasping PCA (Figure 14), actuated conditions generally form clusters distinct from non-actuated trials, indicating that pneumatic assistance alters both neuromuscular and kinematic signatures—though separation varies by participant. The glass-grip PCA (Figure 19) shows similar patterns: some participants' actuated data shift clearly along PC axes, while others overlap, reflecting individual grasp strategies. Finally, the paper-grip PCA (Figure 25) again reveals that actuation shifts sensor profiles for certain participants but not others. Together, these results demonstrate that assistive actuation consistently modifies sensor–feature space, yet the magnitude and direction of those changes are highly participant-specific.

As we aim to get more patients enrolled in the NeuroSuitUp and HEROES studies, we will attempt to investigate the effect of the patients' measured neuropsychological properties on the performance during robotic use and on the neural rehabilitation process. A total of 15 SCI patients and 35 stroke patients are expected to be recruited by the end of the clinical trials, when the development of the final versions of robotic devices is concluded, allowing for meaningful statistical investigations. In this evaluation study, both the SCI and stroke patients demonstrated no apathy, anxiety, depression, or low-self-esteem traits. SCI patients reported a high degree of capacity for mental imagery with VVIQ scores of 80 (0.5) and KVIQ scores of 50 (0). On the other hand, stroke patients reported a decrease in vividness (55 (9)) and in kinesthetic/visual motor imagery (39 (17)) that was, interestingly, uniform with regard to their affected and unaffected hands. The only measurable comparable clinical difference between the two groups can be explored through spasticity levels. The SCI patients scored lower MAS UE levels (1, 1+, 2) in relation to the stroke patients who exhibited higher UE spasticity (MAS 4, 2, 3). The difference was evident during the SRPAS fitting, where it was notably more difficult to fit the device on the spastic-affected hands of the stroke participants. Spasticity has been traditionally documented to be a maladaptive plastic [65] compensation and an important target of mitigation through rehabilitation [66], but newer evidence does not necessarily correlate it to worse functional outcomes [67,68]. The relation of spasticity and poor visualization of movement is even less well understood, but the restriction of movement caused by spasticity should be further investigated as potentially harmful to movement visualization and recovery [69].

The groups of patients that participated in our study commented favorably upon their experience and interaction with the robotics [70,71], both qualitatively and through quantitative scales. Participants experienced very low discomfort levels (most 0 at LED) and reported generally low mental effort (0–10, with only one stroke patient reporting 70 at SMEQ). With regards to their perception of the robotics' characteristics, SCI and stroke participants both reported high Likeability (20 (3) and 17 (6.5), respectively), high perceived intelligence (15 (3.5) and 17 (8), respectively) and maximum perceived safety (15 (0) both groups) at the Godspeed Robotics Questionnaire [72]. They also reported the SRPAS as moderately anthropomorphic (9 (2) and 13 (1.5), respectively), showing moderate animosity (13 (2.5) and (15 (3.5), respectively) and, in general, Godspeed scores comparable to our previous investigations [28,29].

Spearman rank-order correlations (Figure 26) showed that imagery scales (KVIQ, VVIQ) were strongly and positively associated with EMG features—participants with higher imagery vividness exhibited greater mean EMG activity and higher EMG dominant frequencies—whereas Godspeed scores flipped the sign, correlating negatively with EMG but positively with flex peak amplitude, such that better robot perception accompanied larger peak flexion yet lower sustained muscle activity. However, with this limited number of participants, even “perfect” ρ values are almost certainly driven by this binary grouping rather than a continuous, generalizable sensor–questionnaire relationship.

5. Conclusions

The study presented in this manuscript demonstrates that the proposed SRG prototypes can modify sensor-derived features during grasping tasks, ultimately enhancing hand function in individuals with severe motor impairments. In participants with SCI, the SRG facilitated coordinated finger movements by providing consistent and stable pneumatic support. Analyses of finger flexion in the time domain revealed uniform and reliable grip performance across the index, middle, ring, and pinky fingers—a crucial outcome for individuals with compromised motor control.

For the post-stroke participant group, although baseline sensor readings and feature values differed from those in the SCI cohort, the SRPAS similarly induced significant changes in both time- and frequency-domain metrics. Actuation influenced key time-domain features, such as peak amplitude, while frequency-domain measures—namely dominant frequency and total power—revealed changes in the energy distribution of EMG and accelerometer signals. PCA illustrated systematic differences between non-actuated and actuated conditions. Despite greater variability among post-stroke participants, these outcomes underscore the potential of SRGs to modulate neuromuscular and kinematic patterns even in cases of heterogeneous motor deficits.

Overall, the proposed SRGs offer a promising approach to enhance hand function by dynamically altering the sensor profiles associated with grasping tasks. The marked improvements observed in the SCI group, along with the encouraging yet individualized responses in the post-stroke group, highlight the importance of personalized device calibration. We should carefully place the importance of our findings on the feasibility and usability of the SRG under the scope of limiting factors, especially the small number of subjects (three SCI and three stroke patients) that participated in this preliminary study. While some conclusions regarding the promise of the approach can be drawn, no sub-group analysis, including gender analysis, no extensive statistical analysis with regards to injury severity, chronicity, side or level of lesion, and few meaningful comparisons between the two groups of patients can be conducted, except for qualitative observations. Similarly, the current work focused on preliminary analyses of feasibility, usability, and user experience rather than statistical analysis of efficacy and performance against healthy control subjects.

Furthermore, the post-stroke results indicate that although each participant exhibits unique sensor “signatures,” a noticeable shift in feature space occurs between different actuation states—particularly when both time- and frequency-domain features are considered together. Future work of the proposed project’s development will include integration of kinematic and dynamic modeling of human hand and glove complex into the control scheme, use of normative sEMG [73] value ranges [74] for thresholding assistive levels, and integrating and optimizing electrical stimulation in a hybrid actuation scheme for the arm and hand, integrating sensor-derived metrics with clinical performance measures—such as grip strength and task success rates—to further optimize the device’s therapeutic impact across diverse patient populations. More importantly, the conduction and conclusion of the two registered clinical trials, involving SCI and stroke patients, as well as age- and

sex-matched healthy control participants will allow for meaningful statistical analyses with regards to the patient demographics, clinical information, neuropsychological evaluations and extensive performance metrics. Only such analyses will allow us to accurately evaluate and discuss the rehabilitative effect of the platform of devices, and the SRG.

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Data Availability Statement: Data from the NeuroSuitUp and HEROES studies will be made publicly available after the completion of the projects and will be accessible through their respective institutional project web pages under an Attribution-NonCommercial-NoDerivatives 4.0 International license. Data of the analysis presented in the current manuscript and the related code can be made available via a request to the authors following a Memorandum of Understanding (MoU) in the context of Open Research Initiative.

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Abbreviations

The following abbreviations are used in this manuscript:

ADLs	Activities of Daily Living
AES	Apathy Evaluation Scale
BAI	Beck Anxiety Index
BDI	Beck's Depression Inventory
ComSS	Communication Software Subsystem

DoF	Degrees of Freedom
EMS	Electrical Muscle Stimulation
g-SCIM-III	Spinal Cord Independence Measure
IMUs	Inertial Measurement Units
ISNCSCI	International Standard for Neurological Classification of Spinal Cord Injury
KVIQ	Kinesthetic and Visual Imagery Questionnaire
MAS	Modified Ashworth Scale
MSS	Multi-Sensor Subsystem
PCA	Principal Component Analysis
QoL	Quality of Life
ROS	Robot Operating System
RSE	Rosenberg Self-Esteem Scale
SCI	Spinal Cord Injury
sEMG	Surface Electromyography
SR	Soft Robotics
SRG(s)	Soft Robotic Glove(s)
SRPAS	Soft Robotic Pneumatic Actuation Subsystem
VVIQ	Vividness of Visual Imagery Questionnaire

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